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Regression KPI: Can you answer these 20 questions?

Machine Learning Interview Question Part 15

6 min read · Feb 15, 2026



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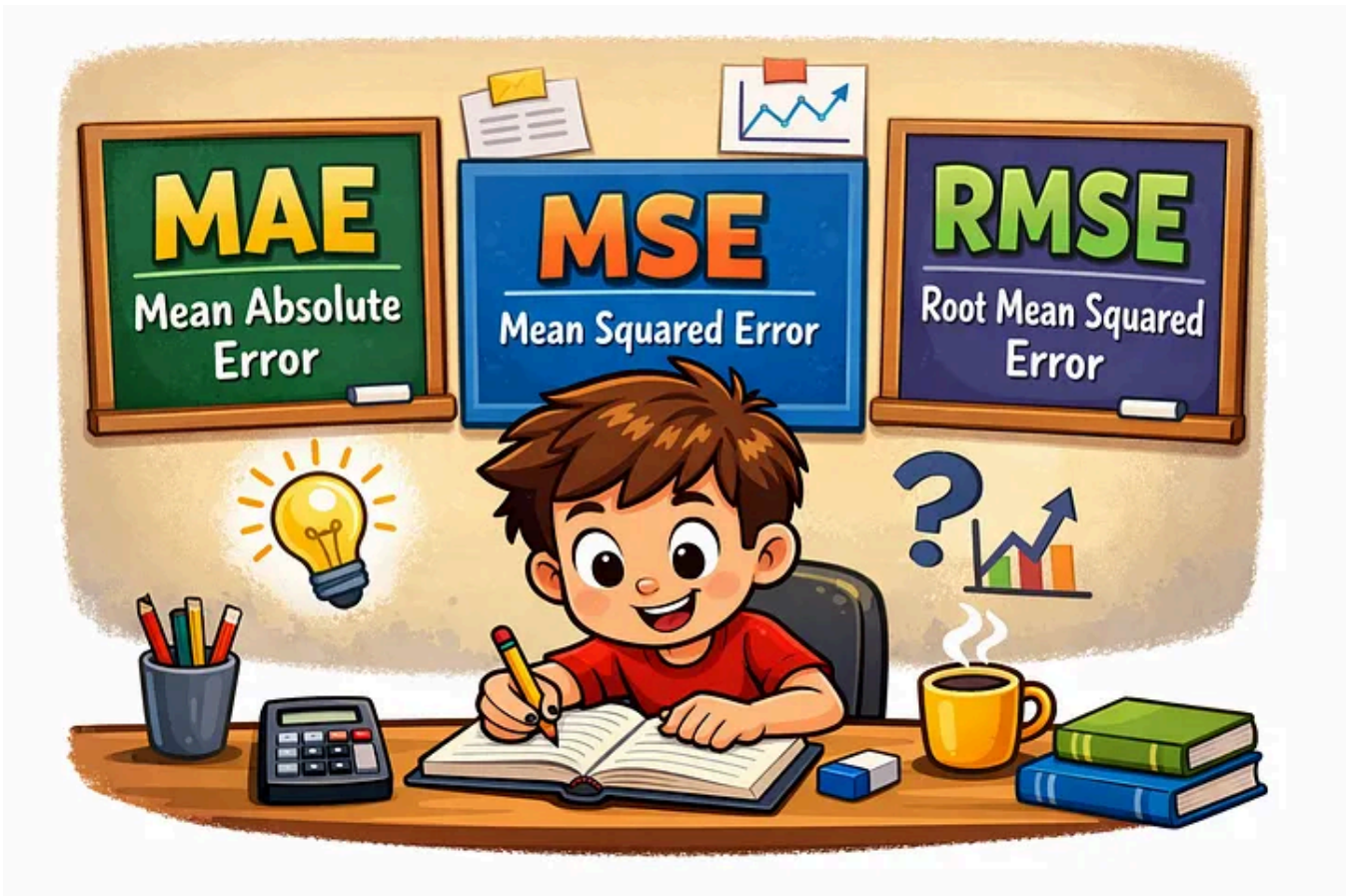
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Crack Data Scientist & Machine Learning Engineer Interview: 400 Questions & Solutions

Key Performance Indicators (KPIs) such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE) provide quantitative ways to measure how closely model predictions align with actual values. Each metric captures error from a different perspective, emphasizing aspects like sensitivity to outliers, interpretability, or scale. Understanding these KPIs is essential for selecting, comparing, and tuning regression models effectively. This blog explores the most common regression error metrics.



Source: Image is generated by ChatGPT

Let's check your basic knowledge of Regression KPI. Here are 20 multiple-choice questions and solutions for you.

1. Mean Absolute Error uses Manhattan distance to calculate the error. A lower MAE indicates a more accurate model. What does MAE measure?

- (A) The average of the squared differences between predicted and actual values.
- (B) The proportion of variance explained by the model.
- (C) The square root of the average of squared differences between predicted and actual values.
- (D) The average of the absolute differences between predicted and actual values.

2. Mean Squared Error uses Euclidean distance to calculate the error. A lower MSE signifies better model performance, indicating that the model's predictions are closer to the true values. What does MSE measure?

- (A) The average of the squared differences between predicted and actual values.
- (B) The proportion of variance explained by the model.
- (C) The square root of the average of squared differences between predicted and

actual values.

(D) The average of the absolute differences between predicted and actual values.

3. What is the primary purpose of Mean Absolute Error (MAE) in regression analysis?

(A) To square the errors before summing them up.

(B) To measure the average magnitude of the errors.

(C) To amplify the effect of outliers in the dataset.

(D) To assess the categorical data accuracy.

4. MAE is less sensitive to outliers than MSE or RMSE because it does not square the errors. What does an MAE of 0 indicate?

(A) The model has high variance.

(B) The model has a high bias.

(C) The model's predictions are perfect.

(D) The dataset contains no outliers.

5. A lower RMSE indicates a more accurate model. Which of the following is true about RMSE?

(A) RMSE is less sensitive to outliers than MSE.

(B) RMSE can be negative.

(C) RMSE is always smaller than MSE.

(D) RMSE is the square root of MSE.

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6. Which metric is most sensitive to outliers in a regression model?

(A) Mean Absolute Error (MAE)

(B) Mean Squared Error (MSE)

- (C) Root Mean Squared Error (RMSE)
- (D) Mean Absolute Percentage Error (MAPE)

7. Which option is correct?

- (A) MSE, due to the squaring of each error, tends to heavily weigh larger errors, making it more sensitive to outliers compared to other metrics.
- (B) Adjusted R-squared is preferred over R-squared in the presence of multiple independent variables as it adjusts for the number of predictors in the model.
- (C) RMSE is in the same units as the target variable, making it more interpretable than MSE.
- (D) All of the options.

8. Which metric would be best for the dataset containing many outliers?

- (A) Mean Absolute Error (MAE)
- (B) Mean Squared Error (MSE)
- (C) R-squared
- (D) Root Mean Squared Error (RMSE)

9. RMSE is the sample standard deviation of the differences between predicted and observed values. In the context of regression analysis, what is the advantage of using RMSE over MSE?

- (A) RMSE is less sensitive to outliers.
- (B) RMSE is easier to interpret in the context of the original data.
- (C) RMSE provides a more conservative estimate.
- (D) RMSE is always lower than MSE.

10. Which metric would be the most appropriate for a regression model with a skewed distribution of errors?

- (A) Mean Squared Error (MSE)
- (B) Root Mean Squared Error (RMSE)

(C) Mean Absolute Error (MAE)

(D) R-squared

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11. Which metric would be the best to use when you have a regression model that is sensitive to the scale of the data?

(A) Mean Squared Error (MSE)

(B) Mean Absolute Error (MAE)

(C) Coefficient of Determination

(D) Root Mean Squared Logarithmic Error (RMSLE)

12. Mean Absolute Percentage Error (MAPE) can become undefined or misleadingly high. When is MAPE not suitable?

(A) The dataset contains zero or near-zero actual values.

(B) The dataset has outliers.

(C) The model is a linear regression.

(D) The dependent variable is continuous.

13. What does R-squared in regression analysis represent?

(A) The ratio of the total sum of squares.

(B) The proportion of the variance in the dependent variable that's explained by the independent variables.

(C) The mean squared error of the model.

(D) The average prediction error.

14. Which metric is known as the coefficient of determination?

(A) Mean Absolute Error (MAE)

(B) Mean Squared Error (MSE)

- (C) R-squared
- (D) Root Mean Squared Error (RMSE)

15. What is the main disadvantage of using R-squared in evaluating regression models?

- (A) It can be negatively affected by outliers.
- (B) It can misleadingly increase with the addition of more predictors, even if they are not relevant to the model, leading to overfitting.
- (C) It measures absolute, not relative, error.
- (D) It does not indicate the direction of the error.

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16. What does an R-squared value of 1 indicate in a regression model?

- (A) No correlation between variables.
- (B) A perfect fit to the data.
- (C) High bias in the model.
- (D) Only a random relationship between variables.

17. Which metric can show a value between $-\infty$ and 1, where 1 indicates a perfect fit?

- (A) Mean Absolute Error (MAE)
- (B) Mean Squared Error (MSE)
- (C) R-squared
- (D) Root Mean Squared Error (RMSE)

18. When Adjusted R-squared is used instead of R-squared?

- (A) There are multiple independent variables.
- (B) The model is a simple linear regression.

- (C) The sample size is very large.
- (D) The model accuracy is extremely high.

19. When Adjusted R-squared is suitable?

- (A) It penalizes the addition of irrelevant predictors.
- (B) It works better with large datasets.
- (C) It simplifies the computation process.
- (D) It eliminates the need for validation data.

20. Which set of metrics are not used to measure the accuracy of a regression model?

- (A) R-squared (higher is better), Adjusted R-squared (higher is better).
- (B) Mean Absolute Error (lower is better), Mean Absolute Percentage Error (lower is better), Median Absolute Error (lower is better).
- (C) Mean Squared Error (lower is better), Mean Squared Log Error (lower is better), Root Mean Squared Error (lower is better), Root Mean Squared Percentage Error (lower is better).
- (D) Accuracy (higher is better), Precision (higher is better), Recall (higher is better), F1 Score (higher is better), ROC-AUC (higher is better).

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The solutions will be found here:

[Top 20 Regression KPI Interview Questions and Answers \(Part 1\)](#)

[Top 20 Regression KPI Interview Questions and Answers \(Part 2\)](#)

Happy learning! Feel free to discuss your thoughts on these questions in the comment section. Don't forget to **clap** and **share** the quiz link with your friends & LinkedIn connections.

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